Extension: Fitting functions to quadratic data

1

a i Linear ii Positive

b i Linear ii Negative

c i Quadratic ii Positive

d i Quadratic ii Negative

e i Quadratic ii Negative

f i Quadratic ii Negative

g i Quadratic ii Positive

h i Quadratic ii Negative

2

a 1

b i No ii No iii No iv Yes

3

a Quadratic, $f(x) = ax^2 + bx + c, a > 0$

b Quadratic, $f(x) = ax^2 + bx + c, a < 0$

4

a 9

b i Yes ii No iii No iv No

Applications

5

a 67.3

b 35

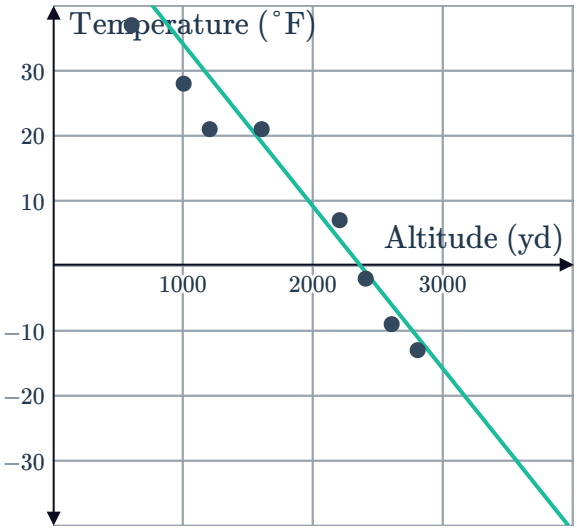
6 **B**

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7

a Linear function

b



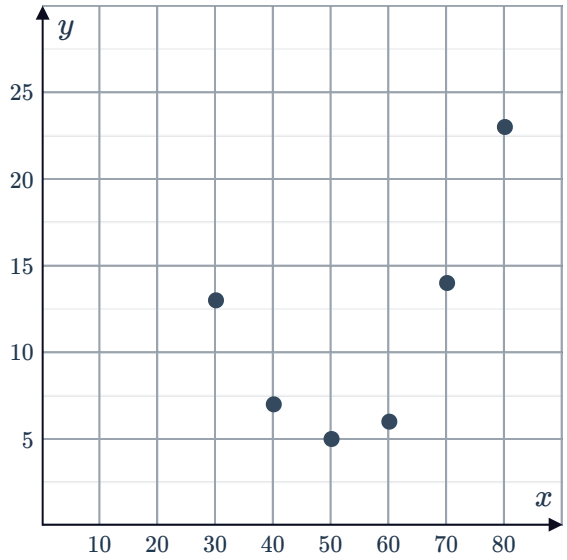
c

Altitude (x)	Average Annual Temperature (y)
500	46.5
1000	34
2000	9

d 36.25°F

8

a



b Quadratic

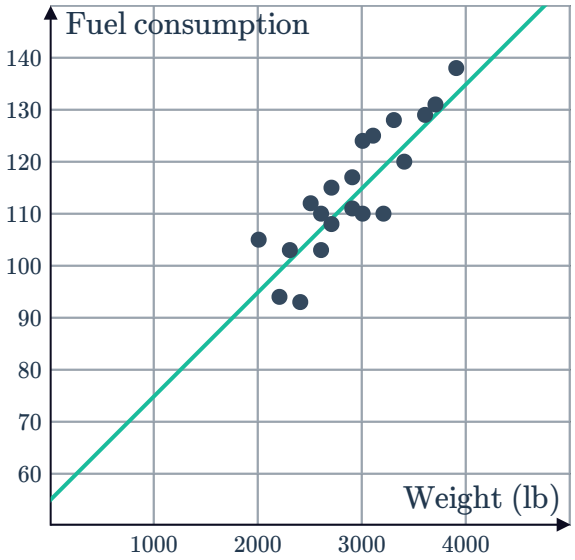
c $y = 0.02x^2 - 2.05x + 56.15$

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9

a Linear function

b



c

Weight (x)	Average Fuel Consumption (y)
3200	119
3100	117
1500	85

d The 1600 lb car

10

a

i 0.25 km

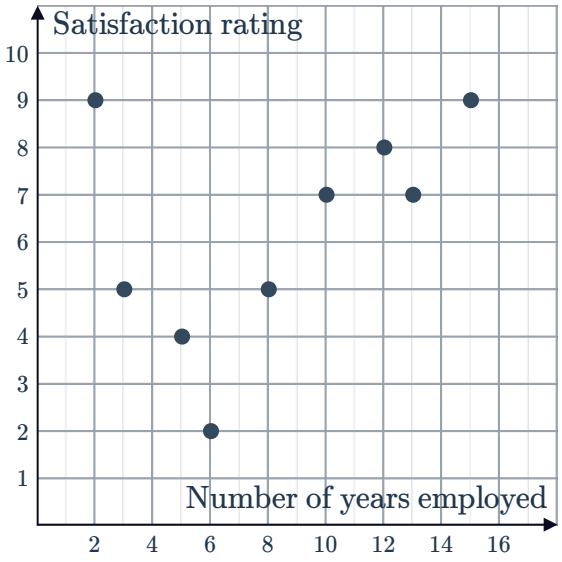
ii 4.375 km

b The extrapolation at $t = 8$ minutes is less reliable since the model predicts that Emma continues to increase her speed for the entire duration of her run.

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11

a



b No

c $y = ax^2 + bx + c$

d 6

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